

Practical 2: Exploring Fourier Space

Objectives

In this practical session we will explore how the Fourier transform connects structures in real space with modes in Fourier space.

The objective is not only to learn how to use the Fast Fourier Transform (FFT) in Python, but also to develop an intuitive understanding of what Fourier modes represent and how manipulating them changes a field in real space.

By the end of the practical you should be able to:

- use the FFT and inverse FFT numerically;
- construct the Fourier frequencies corresponding to a sampled field;
- understand the relation between the size of a real-space domain, its sampling, and the available Fourier modes;
- understand the relation between spatial scales and Fourier frequencies;
- identify the effect of a translation in real space on the Fourier transform;
- understand how sharp and smooth structures appear in Fourier space;
- apply low-pass, high-pass, and other Fourier-space filters;
- extend these ideas from one to two dimensions.

General instructions

The purpose of this practical is not only to obtain the requested numerical results and plots, but also to understand why they have the behavior you observe.

For several parts of the practical, you will be asked to make a prediction **before running your code**. Record your prediction before performing the numerical calculation.

You are allowed to use AI tools; however, you are responsible for understanding every part of the work you submit. During the practical session, the instructors may ask you to explain, justify, or modify any part of your calculation.

1 A localized function in real space

We begin with a simple one-dimensional example.

Consider the function

$$f(x) = \begin{cases} 1, & 3 < x < 4, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

over the interval

$$0 \leq x \leq 7. \quad (2)$$

Create a uniformly sampled array containing N points and evaluate Eq. 1 on this grid.

Plot $f(x)$.

Before computing the Fourier transform

Without performing the FFT, answer:

1. Is $f(x)$ localized or extended in real space? (A localized function is a function whose values are significant mainly within a limited region of its domain and become zero, or very small, outside that region.)
2. Is it a smooth function?
3. Do you expect its Fourier transform to contain only a few modes, or a broad range of Fourier modes?
4. Sketch qualitatively what you expect the amplitude of the Fourier transform to look like.

2 From real space to Fourier space

Compute the Fourier transform of $f(x)$ using

```
np.fft.fft()
```

and construct the corresponding frequency array using

```
np.fft.fftfreq(N, d=dx)
```

where

$$\Delta x = x_{i+1} - x_i. \quad (3)$$

Inspect the array returned by `fftfreq()` before plotting anything.

Questions

1. In what order are positive and negative frequencies stored?
2. What is the smallest non-zero Fourier frequency?
3. What is the largest frequency represented by the grid?
4. What determines the spacing Δk between Fourier modes?
5. What determines the largest $|k|$ that can be represented?

Use

```
np.fft.fftshift()
```

to reorganize the Fourier modes so that zero frequency appears at the center of your plot.

3 The Fourier transform of a top-hat

A top-hat function of width Δ has a Fourier-transform amplitude with a sinc-like envelope,

$$|\hat{f}(k)| \propto \left| \frac{\sin(k\Delta/2)}{k\Delta/2} \right|. \quad (4)$$

For the function in Eq. 1,

$$\Delta = 1. \quad (5)$$

Plot

$$|\hat{f}(k)| \quad (6)$$

and compare its shape with the expected sinc-like behavior.

Questions

1. Does the numerical Fourier transform show the expected oscillatory behavior?
2. Why does the Fourier transform extend to large values of $|k|$?
3. What happens to the Fourier transform if the top hat becomes narrower?
4. What do you expect if the top hat becomes wider?

For questions 4 and 5, make your prediction before modifying the code. Then test it numerically.

4 Amplitude and phase

The Fourier transform can be written as

$$\hat{f}(k) = |\hat{f}(k)|e^{i\phi(k)}, \quad (7)$$

where $|\hat{f}(k)|$ is the Fourier amplitude and $\phi(k)$ is the phase.

Plot separately:

$$|\hat{f}(k)| \quad (8)$$

and

$$\phi(k). \quad (9)$$

Now translate the top-hat function so that its center is located at a different position while keeping its width unchanged.

Before computing the new FFT, predict:

1. Will the Fourier amplitude change?
2. Will the Fourier phase change?

Compute the Fourier transform of the translated function and compare it with the original one. Explain your result.

5 Returning to real space

Apply the inverse Fourier transform using

```
np.fft.ifft()
```

and verify that you recover the original function.

Quantify the reconstruction error using, for example,

$$\epsilon_{\text{rec}} = \max_x |f_{\text{recovered}}(x) - f(x)|. \quad (10)$$

Answer:

1. Is the recovered function exactly equal to the original one within numerical precision?
2. Does the inverse transform contain an imaginary component?
3. If so, how large is it compared with the real component?

6 The role of sampling

Repeat part of the previous analysis using different numbers of real-space samples N , while keeping the physical size of the interval fixed.

Before running the calculation, predict what will happen when N is increased.

Investigate:

1. Does the fundamental Fourier frequency change?
2. Does the maximum available Fourier frequency change?
3. Does the real-space resolution change?

Now keep N fixed and change the physical length of the interval.

Again, predict before calculating:

1. What happens to the spacing between Fourier modes?
2. What happens to the smallest non-zero Fourier frequency?

Summarize the roles of the box size L and sampling interval Δx .

7 Fourier transforms in two dimensions

Fourier transforms can also be applied to two-dimensional fields.

Choose a grayscale image and load it as a two-dimensional array,

$$I(x, y). \quad (11)$$

Display the original image.

Compute its two-dimensional Fourier transform using

```
np.fft.fft2()
```

and use

```
np.fft.fftshift()
```

for visualization.

Because the Fourier amplitude can span a large dynamic range, display

$$\log \left(1 + |\hat{I}(k_x, k_y)| \right). \quad (12)$$

Before looking at Fourier space

Predict:

1. Which parts of the Fourier plane correspond to small-scale image structure?
2. Where do you expect information associated with large-scale variations to appear?

8 Filtering an image

Construct the two-dimensional Fourier-frequency grids k_x and k_y and define

$$k = \sqrt{k_x^2 + k_y^2}. \quad (13)$$

8.1 Low-pass filter

Remove all modes satisfying

$$k > k_0. \quad (14)$$

Before applying the inverse transform, predict what will happen to the image.

Transform back to real space and display the result.

Discuss:

1. Which structures have disappeared?
2. Has the image become smoother or sharper?
3. What happened to the edges?

8.2 High-pass filter

Now remove all modes satisfying

$$k < k_0. \tag{15}$$

Again, predict the result before performing the inverse transform.

Display the resulting image.

Answer:

1. Which structures remain visible?
2. What happened to the large-scale variations in brightness?
3. Why are edges particularly visible after high-pass filtering?

9 Design your own Fourier filter

Design one additional Fourier-space filter.

You are free to decide which modes to modify.

For example, you could:

- retain only an intermediate band of Fourier modes;
- remove a narrow band of frequencies;
- suppress modes predominantly along one direction;
- construct a smooth rather than a sharp cutoff.

Before applying your filter:

1. Describe mathematically which region of Fourier space your filter modifies.
2. Predict the effect on the real-space image.
3. Explain why you expect this effect.

Apply the filter and compare the result with your prediction.

10 What have we learned?

Answer briefly, without running additional calculations:

1. What does a small value of $|k|$ represent in real space?
2. What does a large value of $|k|$ represent?
3. What happens in Fourier space when a function is translated in real space?
4. What determines the Fourier resolution Δk ?
5. What determines the maximum frequency that can be represented?
6. If you know all Fourier amplitudes but discard all Fourier phases, do you still have enough information to reconstruct the original field? Explain.

AI-use statement

At the end of your work, briefly state:

- whether you used an AI tool during the practical;
- what you used it for;
- one suggestion or result obtained from the AI that you independently checked;
- how you verified it.

What should you submit?

Submit your notebook or code together with a short report containing:

- your predictions made before the numerical calculations;
- the requested real-space and Fourier-space plots;
- your FFT/IFFT consistency check;
- your investigation of amplitude and phase;
- your Fourier-filtering experiments;
- your investigation of sampling and Fourier resolution;
- your two-dimensional image analysis;
- your own Fourier filter and its interpretation;
- your AI-use statement.

The practical will be evaluated not only from the final plots, but also from your ability to explain the relation between Fourier modes and structures in real space.